

Q3LOCK Second Weighted-Energy Moment and Common-Alpha Cauchy Gate

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This note records [EXP-000794](#) and reusable result [R-167](#). It asks how far the exact fixed-lattice ST8/Q3LOCK Hamiltonian can be pushed from the first weighted-energy cone toward one phase-independent thermodynamic real-time dynamics.

There is a strict positive advance. The exact energy current closes the second weighted-energy moment, the minimal three-half-energy moment, propagation of the energy domain $D(A^{3/4})$, and the boundary position multiplier. It does not prove the noncommuting cubic multiplier or the corresponding graph-domain embedding. An exact two-sided Duhamel theorem then reduces thermodynamic Cauchy to one explicit spatial commutator gate.

There is also a strict route correction. Four plausible shortcuts fail by exact counterexamples: automatic squaring of positive form order, symmetric energy sandwiches as a substitute for strong convergence, polynomial separate-rung energy conjugation, and convexity alone as a weighted commutator sign. The surviving proof target is product-level rather than rung-by-rung.

No common thermodynamic automorphism, common-alpha KMS identification, algebraic ground state, GNS or physical mass gap, continuum removal, physical empty-space comparison, C6, CP1, Sector A or Pre-A closure is claimed.

2 Provenance identifiers

- Result: [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#).
- Scope correction: [EXP-000795](#); it narrows the issued [EXP-000794](#) cubic-domain wording to energy-domain propagation.
- Next gate: [PA-CP1-ST8-Q3LOCK-ENERGY-WEIGHTED-COMMUTATOR-GEVREY-LR-CLOSURE](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-FIRST-MOMENT-AUTOMATIC-POWER-UPGRADE](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-SYMMETRIC-SANDWICH-ONLY-THERMODYNAMIC-CAUCHY](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-POLYNOMIAL-ALL-RUNG-ONSITE-ENERGY-CONJUGATION](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-CONVEXITY-ONLY-WEIGHTED-COMMUTATOR-SIGN](#).

3 Exact model and first-moment input

At every site $x \in \mathbb{Z}^3$, let $q_x, p_x \in \mathbb{R}^8$. The finite-volume Hamiltonian is

$$H_\Lambda(h) = \sum_{x \in \Lambda} \left\{ \frac{|p_x|^2}{2\chi} + \frac{r}{2}|q_x|^2 + W_4(q_x) - hu \cdot q_x \right\} + \frac{c}{2} \sum_{\langle xy \rangle} |q_x - q_y|^2, \quad (2.1)$$

where $\chi, c, g > 0$, $\lambda \geq 0$, and

$$W_4(q) = \frac{g}{4} \sum_e q_e^4 + \frac{\lambda}{4} \sum_{e \sim f \text{ in } Q_3} (q_e - q_f)^2 (q_e^2 + q_f^2). \quad (2.2)$$

For $|h| \leq h_0$ and $0 < \gamma < g/32$, a scalar shift makes the local energies positive with

$$e_x \geq \frac{|p_x|^2}{2\chi} + \gamma|q_x|^4 + \frac{c}{4} \sum_{y \sim x} |q_x - q_y|^2. \quad (2.3)$$

Fix $f_x = e^{-\mu d(x, X)}$, $S_f = \sum_x f_x < \infty$, and set

$$A = 1 + E_f = 1 + \sum_x f_x e_x = T_f + U_f + 1. \quad (2.4)$$

The first-moment theorem from EXP-000792 is

$$U_\Lambda(t)^* A U_\Lambda(t) \leq e^{v_\mu |t|} A, \quad v_\mu = 6\sqrt{\frac{c}{2\chi}} (e^\mu - 1). \quad (2.5)$$

This controls only one energy moment. Positive operator order is not preserved by taking powers, so (2.5) cannot simply be squared.

4 Why automatic squaring fails

Let

$$E = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \quad U = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad A_1 = U^* E U, \quad c_0 = \frac{5}{2}. \quad (3.1)$$

Exact rational arithmetic gives

$$(c_0 E - A_1)_{11} = \frac{11}{50}, \quad \det(c_0 E - A_1) = \frac{7}{20} > 0, \quad (3.2)$$

but

$$\det(c_0^2 E^2 - A_1^2) = -\frac{127}{16} < 0. \quad (3.3)$$

Thus the first form cone is not a higher-moment theorem. The next section uses the exact Q3LOCK current rather than this invalid order inference.

5 Exact second weighted-energy moment

On the common Schwartz/form core let

$$B_f = \delta(E_f) = \frac{i}{\hbar} [H_\Lambda(h), E_f]. \quad (4.1)$$

The local continuity equation gives

$$B_f = - \sum_{\langle xy \rangle} (f_x - f_y) \frac{c}{2\chi} (p_x + p_y) \cdot (q_x - q_y). \quad (4.2)$$

Put $\theta = e^\mu - 1$ and $z = 6$. Weighted Cauchy–Schwarz yields

$$\|B_f \psi\|^2 \leq \frac{cz\theta^2}{\chi^2} \int U_f \sum_x f_x |p_x \psi|^2. \quad (4.3)$$

The remaining term has the exact Schroedinger cross identity

$$\int U_f \sum_x f_x |p_x \psi|^2 = 2\chi \Re \langle T_f \psi, U_f \psi \rangle + \frac{\hbar^2}{2} \int \left(\sum_x f_x \Delta_x U_f \right) |\psi|^2. \quad (4.4)$$

For one Q3 edge with coordinates a, b ,

$$(\partial_a^2 + \partial_b^2) \frac{\lambda}{4} (a - b)^2 (a^2 + b^2) = \lambda (4a^2 - 6ab + 4b^2) \leq 7\lambda (a^2 + b^2). \quad (4.5)$$

Since every Q3 vertex has degree three,

$$\Delta W_4(q) \leq C_2 |q|^2, \quad C_2 = 3g + 21\lambda. \quad (4.6)$$

For every $\epsilon > 0$,

$$\sum_x f_x \Delta_x U_f \leq C_2 \epsilon U_f + S_f \left(8r_+ + 8cz + \frac{C_2}{4\epsilon\gamma} \right). \quad (4.7)$$

Equations (4.3)–(4.7) close both operator orientations:

$$\|B_f A^{-1}\| = \|A^{-1} B_f\| \leq M_\mu, \quad (4.8)$$

where

$$M_\mu = \theta \left\{ \frac{cz}{\chi} \left[1 + \frac{\hbar^2}{2\chi} \left(C_2 \epsilon + S_f \left(8r_+ + 8cz + \frac{C_2}{4\epsilon\gamma} \right) \right) \right] \right\}^{1/2}. \quad (4.9)$$

Because $\delta(A^2) = AB_f + B_f A$, form Gronwall gives

$$U_\Lambda(t)^* A^2 U_\Lambda(t) \leq e^{2M_\mu |t|} A^2, \quad \|AU_\Lambda(t) A^{-1}\| \leq e^{M_\mu |t|}. \quad (4.10)$$

The constants are uniform in finite volume, sign phase, and the admitted compact source interval.

6 Three-quarter energy-domain propagation

Interpolating (2.5) and (4.10) gives

$$\|A^{3/4}U_\Lambda(t)A^{-3/4}\| \leq \exp \left[\left(\frac{v_\mu}{4} + \frac{M_\mu}{2} \right) |t| \right], \quad (5.1)$$

or equivalently

$$U_\Lambda(t)^* A^{3/2} U_\Lambda(t) \leq e^{(v_\mu/2 + M_\mu)|t|} A^{3/2}. \quad (5.2)$$

Scalar power counting identifies a necessary target for the cubic force: in the commuting model $A \sim 1 + \gamma q^4$, $q^3 A^{-s}$ can be bounded only for $s \geq 3/4$. A symmetric sandwich analog needs $s \geq 3/8$. This power count does not prove $q^3 A^{-3/4}$ bounded for the noncommuting anharmonic Schroedinger energy, nor the embedding $D(A^{3/4}) \subset D(q^3)$. Equations (5.1)–(5.2) prove energy-domain propagation only; the cubic multiplier and domain embedding remain part of the next gate.

The boundary position multiplier also closes. Balakrishnan calculus gives

$$\|[A^{1/2}, q_x]A^{-3/4}\| \leq \hbar \sqrt{\frac{f_x}{2\chi}}, \quad (5.3)$$

and therefore

$$f_x^{1/4} \|A^{1/2} q_x A^{-3/4}\| \leq \gamma^{-1/4} + \frac{\hbar}{\sqrt{2\chi}}, \quad (5.4)$$

with the adjoint orientation as well.

7 Conditional thermodynamic-Cauchy theorem

For $\Lambda \subset \Lambda'$, the exact boundary Duhamel identity is

$$\tau_t^{\Lambda'}(A_0) - \tau_t^\Lambda(A_0) = \frac{i}{\hbar} \int_0^t \tau_{t-s}^{\Lambda'} [V_{\partial\Lambda}, \tau_s^\Lambda(A_0)] ds. \quad (6.1)$$

For $R = q_x - q_y$, $C_1^a = [R_a, D]$, and $C_2^a = [R_a, C_1^a]$,

$$[c|R|^2/2, D] = \frac{c}{2} \sum_a (C_2^a + 2C_1^a R_a). \quad (6.2)$$

Assume both one-sided orientations satisfy, uniformly in finite volume and compact time,

$$\|C_1^a A^{-1/2}\| + \|A^{-1/2} C_1^a\| \leq C e^{-\rho(d(x,X)-v|t|)}, \quad (6.3)$$

$$\|C_2^a A^{-3/4}\| + \|A^{-3/4} C_2^a\| \leq C e^{-\rho(d(x,X)-v|t|)}, \quad \rho > \frac{\mu}{4}. \quad (6.4)$$

Then the cubic-lattice shell sum and (5.1)–(5.4) give exponential boundary decay in both one-sided $A^{-3/4}$ -damped norms. Union comparison makes the limit exhaustion independent. The two orientations pass products, adjoints, group law, inverse, and strong time continuity on the stable core. An invariant Hamiltonian-derived C-star algebra or norm-tight energy tails are still required for the C-star conclusion.

Symmetric damping alone is insufficient. On $\ell^2(\mathbb{N}_0)$, let $W e_n = (n+1)e_n$ and $D_n = |e_n\rangle\langle e_0|$. Then both symmetric sandwich norms are $(n+1)^{-s} \rightarrow 0$, including the adjoints, but $D_n e_0 = e_n$ is not strongly Cauchy.

8 The infinite ladder and the rejected rung route

The cubic force does not preserve a finite first/double commutator system:

$$[q^3, D] = 3[q, D]q^2 + 3[q, [q, D]]q + [q, [q, [q, D]]]. \quad (7.1)$$

A proposed repair was a polynomial separate-rung estimate for $K^{j/2}V(t)K^{-(j+1)/2}$. It is exactly false. Let $k\phi_m = \epsilon_m\phi_m$ be a positive confining onsite Hamiltonian, $K = k \otimes 1 + 1 \otimes k$, and $V = -c \sum_a q_a \otimes q_a$. A nonzero upward transition gives

$$\|K^{j/2}VK^{-(j+1)/2}\| \geq \frac{b_n}{\sqrt{2\epsilon_0}} \left(\frac{\epsilon_n}{\epsilon_0}\right)^{j/2}, \quad b_n = c \sum_a |\langle \phi_n, q_a \phi_0 \rangle|^2 > 0. \quad (7.2)$$

The bound is unchanged in the onsite interaction picture. In the exact fixture

$$K = \text{diag}(1, 4), \quad V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (7.3)$$

the norm is 2^j . No fixed polynomial in j can dominate it. A simple exponential rung allowance is also unsuitable for sequential multiplication, which produces an $\exp[O(n^2)]$ cost.

The base bond rung survives:

$$\|V_{xy}(t)K_{xy}^{-1/2}\|, \|K_{xy}^{-1/2}V_{xy}(t)\| \leq \frac{c}{\sqrt{2\gamma}}. \quad (7.4)$$

The replacement must estimate the complete Volterra product without moving a growing power through every bond. Candidate architectures are a direct linked-cluster estimate, a heat/strip-loss analytic ideal, or a KMS-specific state-weighted theorem, followed by a noncollapsing finite-density algebra bridge.

9 Convex subregime: a true lemma and a false promotion

For $0 \leq \lambda \leq 2g$, the quartic is genuinely convex and

$$D^2W_4(q)[\xi, \xi] \geq 3(g - \lambda/2) \sum_a q_a^2 \xi_a^2. \quad (8.1)$$

This gives an unweighted Hilbert–Schmidt monotonicity sign, but not the energy/KMS-weighted operator sign needed in (6.3)–(6.4). Already for a scalar convex quartic set $q = \text{diag}(0, 1, 2)$, let D be the negative all-ones matrix, $C = [q, D]$, and $F = [q^3, D]$. Then

$$X = \frac{C^*F + F^*C}{2} = \begin{pmatrix} 17 & 11 & -4 \\ 11 & 8 & 5 \\ -4 & 5 & 23 \end{pmatrix}. \quad (8.2)$$

Although $\text{tr } X = 48$, the vector $v = (-2, 2, -1)$ satisfies $v^*Xv = -1$. Hence the faithful weight $(vv^* + \epsilon I)/(9 + 3\epsilon)$ has negative expectation for $0 < \epsilon < 1/48$. Convexity alone cannot provide the missing weighted commutator estimate.

10 Evidence map and next gate

The successful result is

R-167: exact second weighted-energy moment and Cauchy split.

The four registered route boundaries are:

1. first-moment automatic power upgrade;
2. symmetric-sandwich-only thermodynamic Cauchy;
3. polynomial all-rung onsite-energy conjugation;
4. convexity-only weighted commutator sign.

The next gate is

PA-CP1-ST8-Q3LOCK-ENERGY-WEIGHTED-
COMMUTATOR-GEVREY-LR-CLOSURE.

Its accepted target is no longer the rejected polynomial rung estimate. It must prove product-level spatial decay in both one-sided norms, or an exact state-weighted substitute, and then construct the phase-independent Hamiltonian-derived algebra. Only after that gate closes can the programme identify the two phasewise systems as KMS states of one common real-time dynamics and proceed honestly to ground-state selection and GNS-gap tests.

11 Devil’s-advocate audit

1. **Objection: first-moment form order can simply be squared. UPHELD against that route.** The exact fixture (3.1)–(3.3) has positive first order and indefinite squared order.
2. **Objection: a symmetric energy sandwich is enough for an automorphism. UPHELD against that route.** The rank-one shift sequence is symmetric-sandwich small but not strongly Cauchy.
3. **Objection: the second moment closes the spatial hierarchy. UPHELD as false.** Equation (7.1) creates the third commutator and the ordered alternative creates an infinite graph ladder.
4. **Objection: a polynomial separate-rung bound supplies the missing ladder. UPHELD against that route.** The spectral transition lower bound is exponential in the rung and is unchanged by onsite interaction-picture time evolution.
5. **Objection: quartic convexity closes the weighted sign. UPHELD against that promotion.** The convexity lemma is true, but the faithful weighted 3-by-3 fixture has negative expectation.
6. **Objection: the route failures prove common dynamics does not exist. DISMISSED.** Product-level cancellations, heat-loss ideals, state-weighted methods and other Hamiltonian-derived topologies remain open.

12 Verification and no-overclaim boundary

The primary SymPy implementation derives the Q3 graph, edge Laplacian, second-moment constants, order counterexample, free noncommutative cubic identity, spectral-rung fixture, and convexity-weighted fixture. A separate standard-library Fraction implementation reconstructs the same exact core without importing the primary. The integrated verifier reruns both in fresh temporary locations, compares their derived invariants, checks source hashes, and requires this note, its rendered PDF, the exploration record, result, negative entries, live gate and T-054 route.

This checkpoint does not select a physical functional or a vacuum. It does not prove the cubic multiplier/domain embedding, a common alpha, KMS identification, ground-state purity, broken-sector GNS gap, regulator or continuum limit, physical empty-space reference, below-empty sign, light, mass, time, gravity, C6, CP1, Sector A or Pre-A. Those claims remain gated.

13 Reproduction

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_second_weighted_energy_cauchy_gate.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_second_weighted_energy_cauchy_gate_
independent.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_second_weighted_energy_cauchy_gate_
verify.py
```

Run the primary and independent implementations first and the integrated freshness/provenance verifier last.

14 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167.
Exploration:	EXP-000794; scope correction EXP-000795.
Precise statement:	Exact second energy moment, minimal $A^{(3/2)}$ moment, $D(A^{(3/4)})$ energy-domain propagation and boundary multiplier proved; the cubic multiplier embedding remains open; thermodynamic Cauchy is reduced to a two-sided spatial commutator hypothesis.
Scope:	T0 finite-volume analytic theorem and conditional Cauchy reduction.
Dependencies:	EXP-000792--EXP-000794; the exact fixed-volume ST8/Q3LOCK Hamiltonian and first energy cone.
Evidence grade:	ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:	Run the three commands under Reproduction in the listed order, with the integrated verifier last.
Expected output:	Primary PASS 51/51; independent PASS 271/271; integrated PASS 176/176.
Falsification gate:	Any Q3, current, Laplacian, graph exponent, one-sided topology, spectral-rung, convexity-sign, hash, scope or provenance mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	Cubic multiplier/domain embedding, common alpha/KMS, algebraic ground states, GNS gap, continuum, physical empty, C6, CP1, Sector A and Pre-A remain open.
Next required action:	Prove the cubic multiplier/domain obligation and product-level spatial decay, then construct the Hamiltonian-derived phase-independent algebra.